

Electrical Safety

References :

- Lessons in Electric Circuits, Ch. 3:
http://www.ibiblio.org/kuphaldt/electricCircuits/DC/DC_3.htm
- **Class notes**

Effect of Electricity on Living Tissue

- **Electric "shock"**: Electricity causes our body to experience pain or trauma
 - If fortunate → tingles or jolts of pain

Two main effects:

1. Current heats up the tissue

→ tissue may be burnt (well beneath the skin, i.e., organs)

2. Nervous system ("nerve cells"):

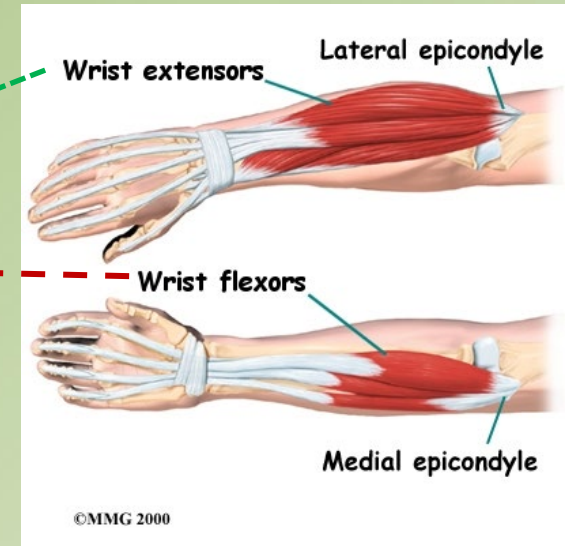
→ Nerve cells communicate using small electric currents/voltages

→ *e.g., Muscles triggered by an external (shock) current will involuntarily contract, and there's nothing the victim can do about it.*

Effect on the nervous system

<http://www.eorthopod.com/content/elbow-anatomy>

- **Tetanus** → *involuntary muscle contraction*
 - forearm muscles for **bending** fingers overcome muscles for **extending** them
 - if electric **current** passes through a person's arm
 - "bending" muscles will win, **clenching the fingers** into a fist.
 - immobilized victim "**frozen on the circuit**"



... *Effect on the nervous system*

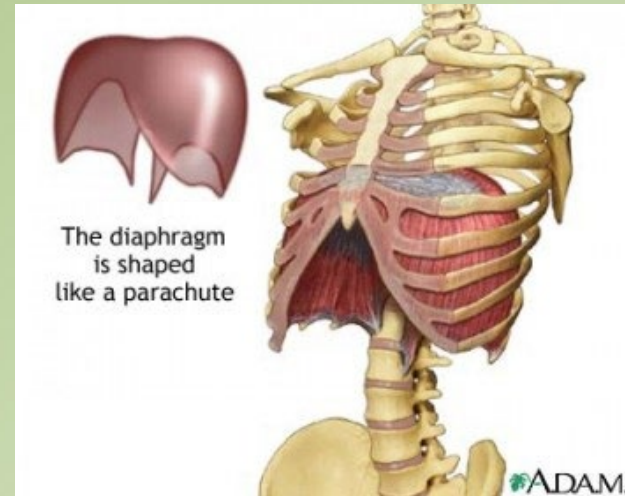
Tetanus can only be interrupted by *stopping the current* through the victim

→ *the effect may stay for a while*

e.g., **Tasers**: momentarily shock a victim with a high-voltage to immobilize him/her (for a few minutes).

... *Effect on the nervous system*

- **Diaphragm muscle** controlling the **lungs** and **heart**
 - heart cannot beat properly
 - a condition known as **fibrillation** (*can result in death*)
- **AC: Low-frequency (50-60Hz) can be**
 - *more dangerous than high-frequency AC*
 - *3 to 5 times more dangerous than DC of the same voltage/current.*



... Effect on the nervous system

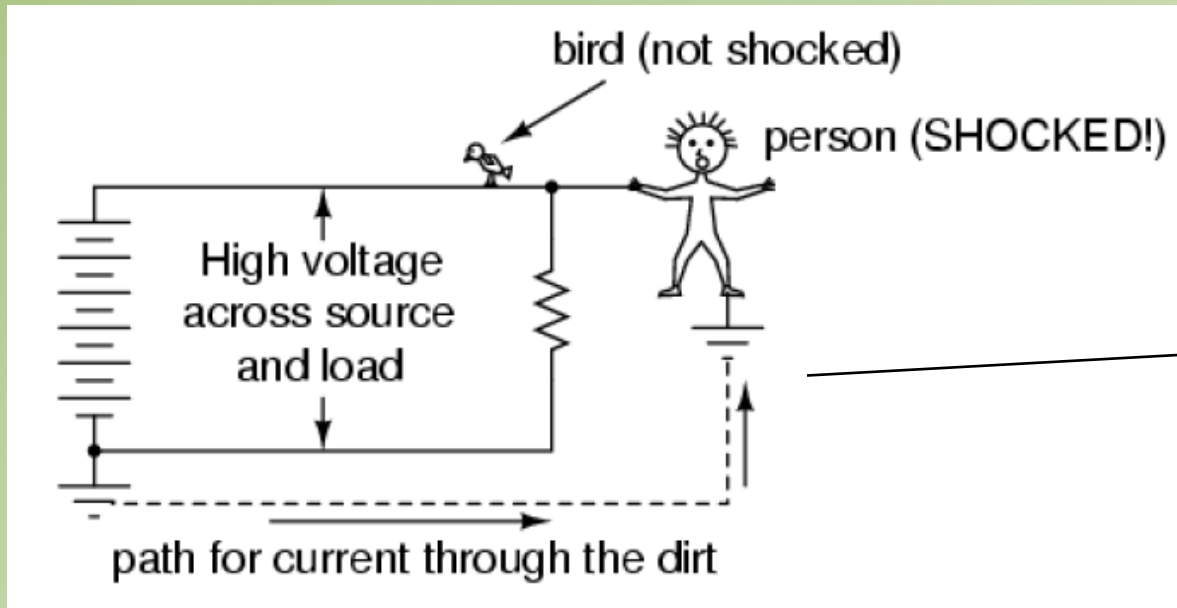
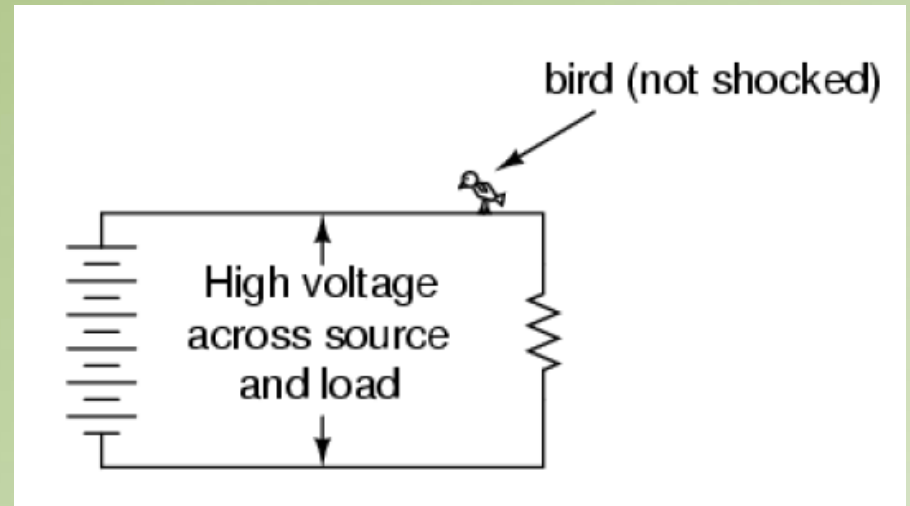
AC is more likely to cause a victim's heart to fibrillate → more dangerous condition after shocking current has been halted.

DC: More likely to "freeze" a victim in a shock scenario → tetanus

**Which is more dangerous?
AC or DC**

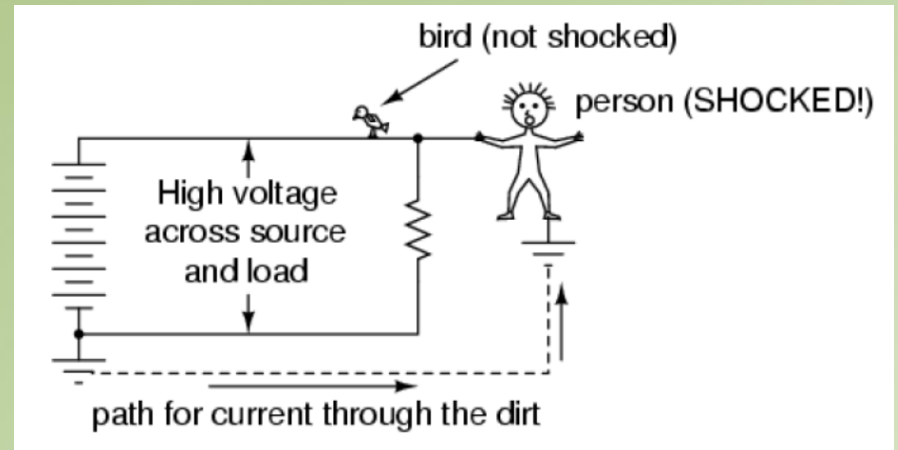
Shock Current Path

Electricity requires a complete path (circuit) to continuously flow



Presence of a ***ground point*** provides an easy point of contact for someone to get shocked

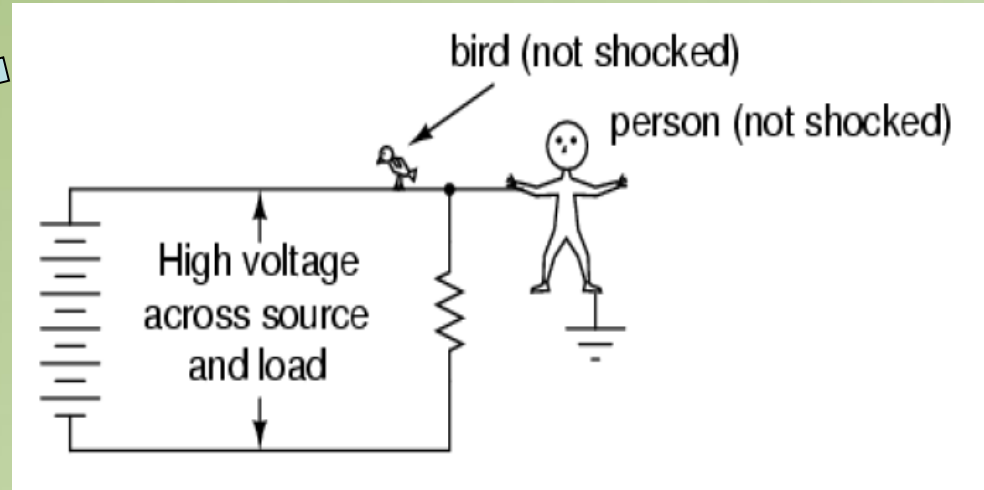
... Shock Current Path:



... Shock Current Path

Q: Is a ground-less circuit safer?

- *intentional "grounding" is to ensure that at least one side is safe*



- **Can we leave the circuit completely un-grounded?**

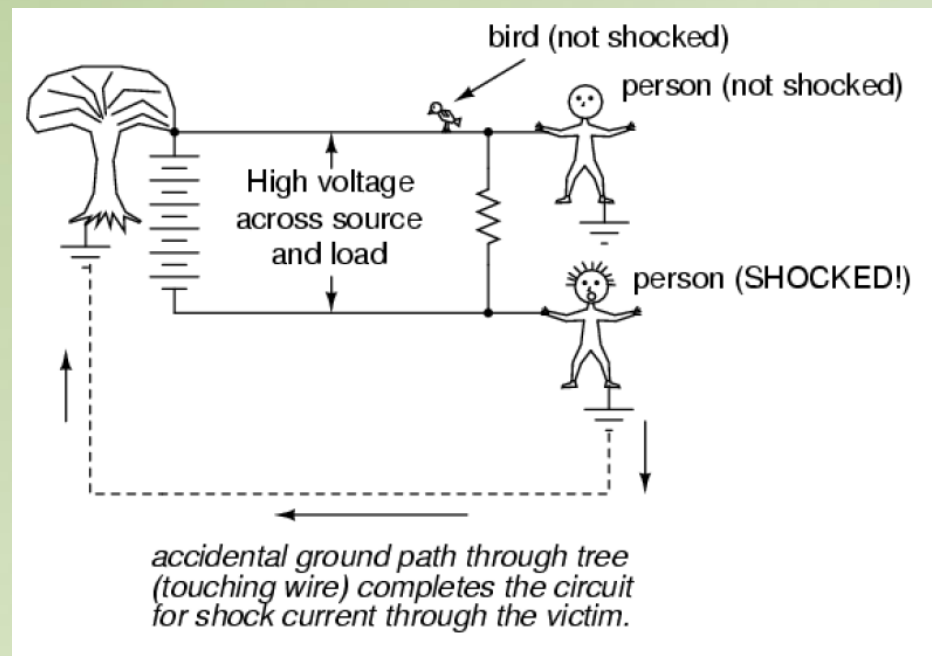
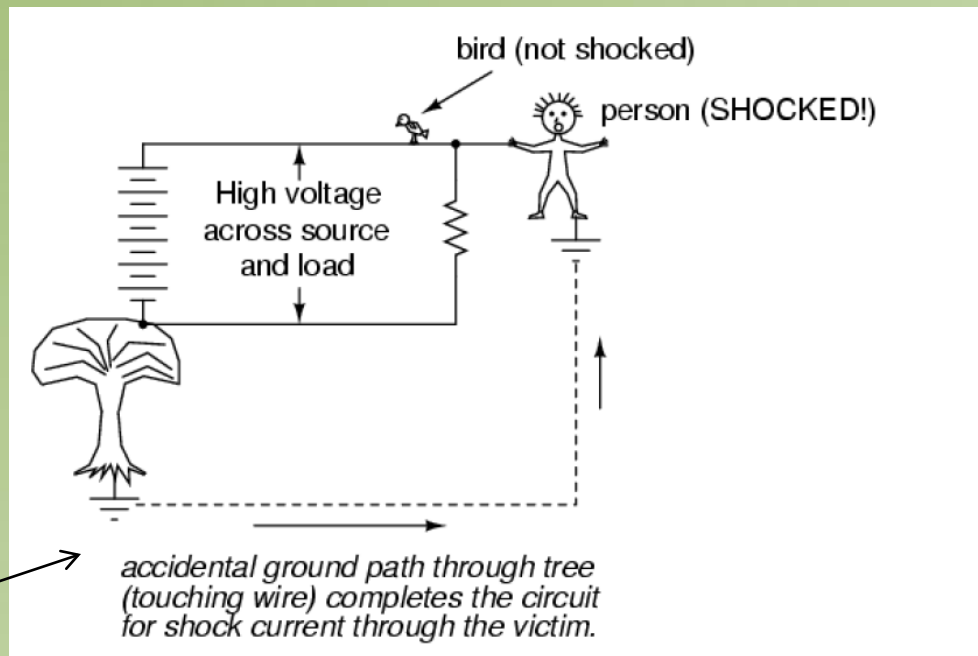
- *Accidental grounds*
→ next

Accidental grounds

A tree branch can cause an accidental ground (ground fault)

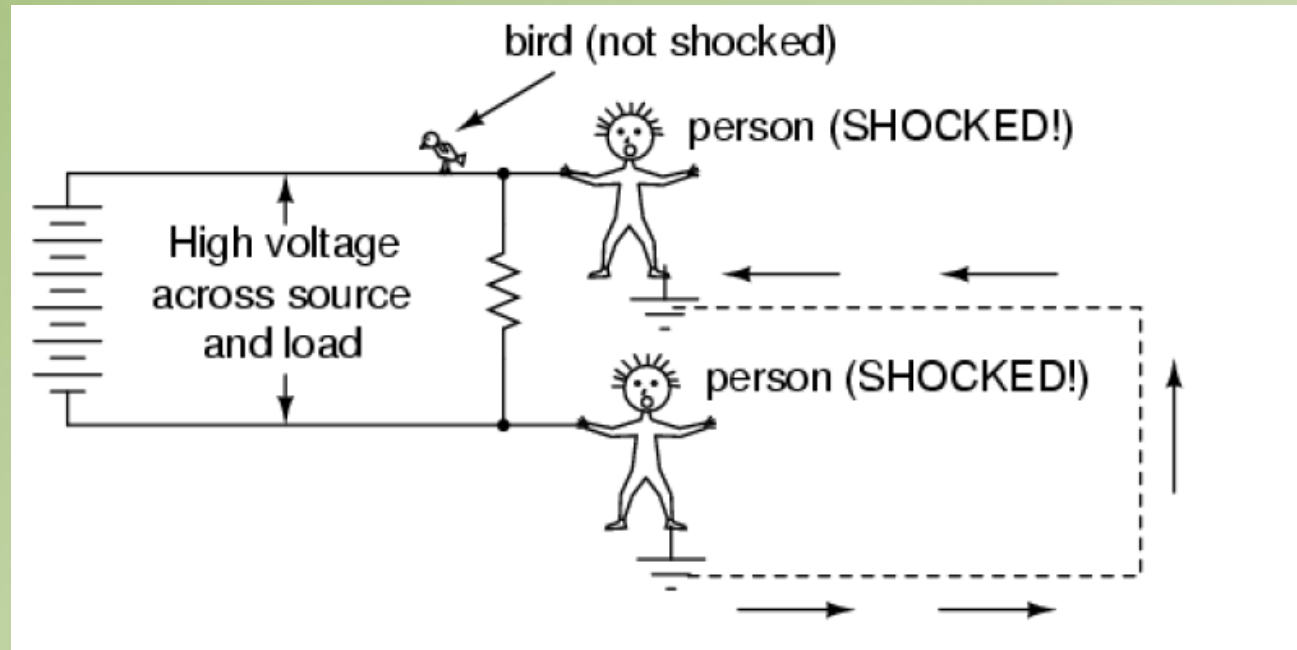
Depending on where he/she is, it can shock or not shock the person

➔ *Unpredictable*



In this example, 2 persons are touching the wire at the same time

Accidental grounds

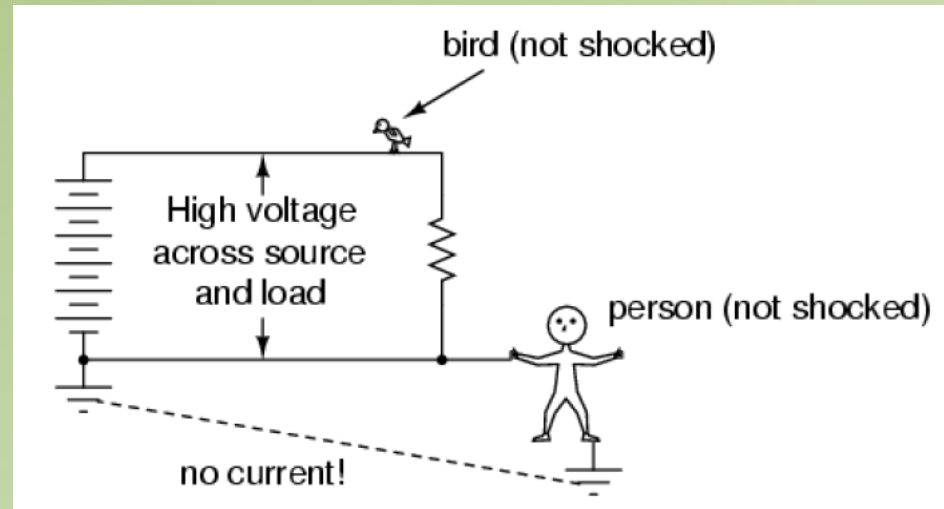


- ***The only character guaranteed to be safe in these scenarios is → the bird → no connection to earth)***
- ***By firmly connecting a designated point to earth ground ("grounding" the circuit), at least safety can be assured at that one point.***

... Shock Current Path:

Q. Can the shoes protect the person by preventing a circuit?

- **Rubber-soled** shoes do indeed provide some electrical insulation to help protect someone from conducting shock current through their feet.
- However, **most common shoe designs** are **not** intended to be electrically "**safe**" → soles are too thin and not of the right substance.

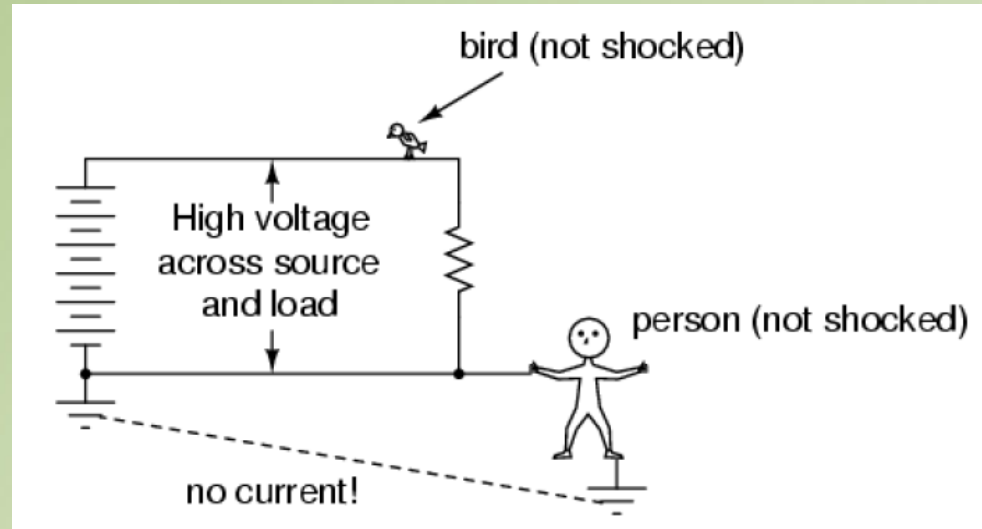


- Hand or foot contact, insulated with **rubber**: 20 MΩ typical
- Foot contact through **leather** shoe sole (**dry**): 100 kΩ to 500 kΩ
- Foot contact through **leather** shoe sole (**wet**): 5 kΩ to 20 kΩ

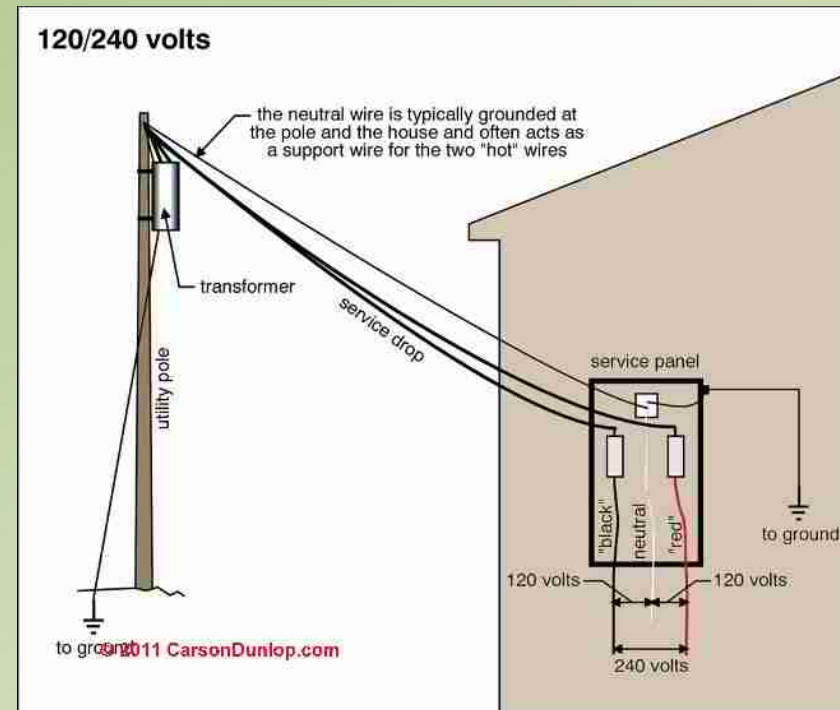
... Shock Current Path:

Q. How good a conductor is dirt?

- *dirt is not a very good conductor*
- *However, it takes very little current to injure or kill a human being*
- *Poor conductivity of dirt is enough to provide a path for deadly current when there is sufficient voltage available*



- Power circuits usually have a designated point that is “***grounded***”:
 - firmly connected to metal rods or plates buried in the dirt to ensure that one side of the circuit is always at ground potential (zero voltage between that point and earth ground)
- Neutral Wire → white in Canada/US
- Protective ground wire, no insulation or GREEN



“It's not voltage that kills, it's current!”

- Electric current: *There must be voltage available to motivate electrons to flow through a victim*
 - *Ohm's law*
- *Body resistance (e.g., $1\text{M}\Omega$)*
 - *not a fixed quantity*
 - *varies from person to person, time to time.*
 - *depends on how the contact is made (area, sweat, ...)*
- *How much current is harmful?*

... It's not voltage that kills, it's current!"

How much current is harmful?

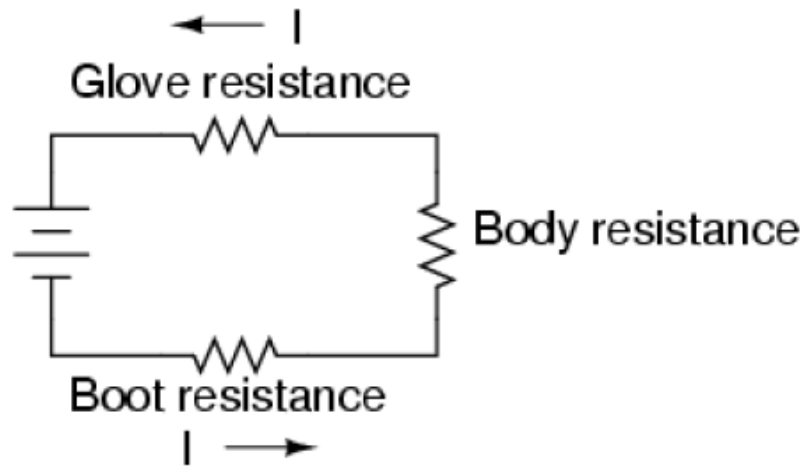
BODILY EFFECT	DIRECT CURRENT (DC)	60 Hz AC	10 kHz AC
Slight sensation felt at hand(s)	Men = 1.0 mA Women = 0.6 mA	0.4 mA 0.3 mA	7 mA 5 mA
Threshold of perception	Men = 5.2 mA Women = 3.5 mA	1.1 mA 0.7 mA	12 mA 8 mA
Painful, but voluntary muscle control maintained	Men = 62 mA Women = 41 mA	9 mA 6 mA	55 mA 37 mA
Painful, unable to let go of wires	Men = 76 mA Women = 51 mA	16 mA 10.5 mA	75 mA 50 mA
Severe pain, difficulty breathing	Men = 90 mA Women = 60 mA	23 mA 15 mA	94 mA 63 mA
Possible heart fibrillation after 3 seconds	Men = 500 mA Women = 500 mA	100 mA 100 mA	

Approximate
figures

Most data regarding induced fibrillation come from animal testing:

- It has been suggested that an across-the-chest current of only **17 milliamps** AC is enough to induce fibrillation in a human subject*
- In industry, **30V** is generally considered to be a conservative threshold value for dangerous voltage.*
- Keep hands clean and dry, and remove all metal jewelry when working around electricity.*
- Use one hand (right hand) to work on a circuit*

- *The best protection against shock from a live circuit is resistance*
 - *through the use of insulated tools, gloves, boots, and other gear.*

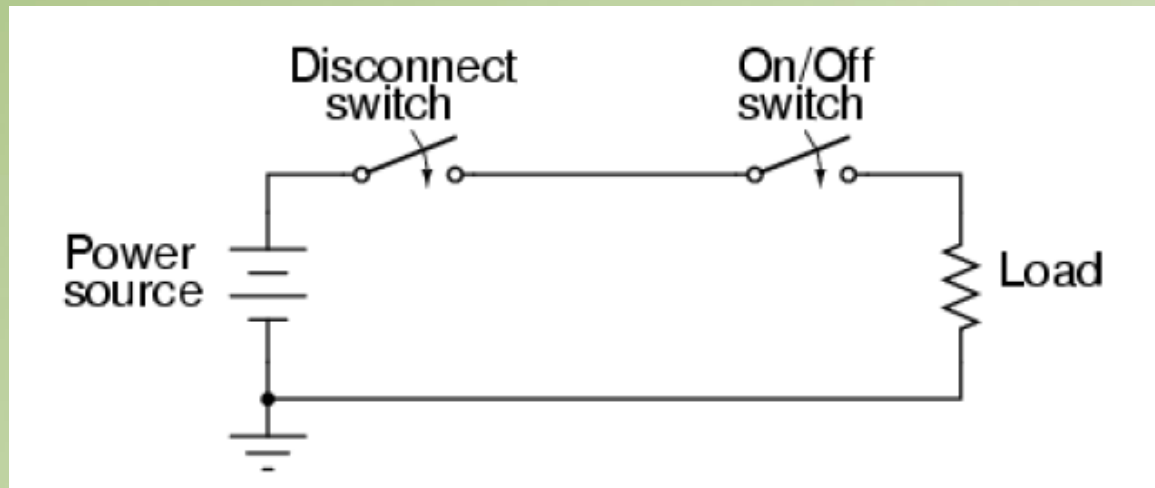


$$I = \frac{E}{R_{\text{glove}} + R_{\text{body}} + R_{\text{boot}}}$$



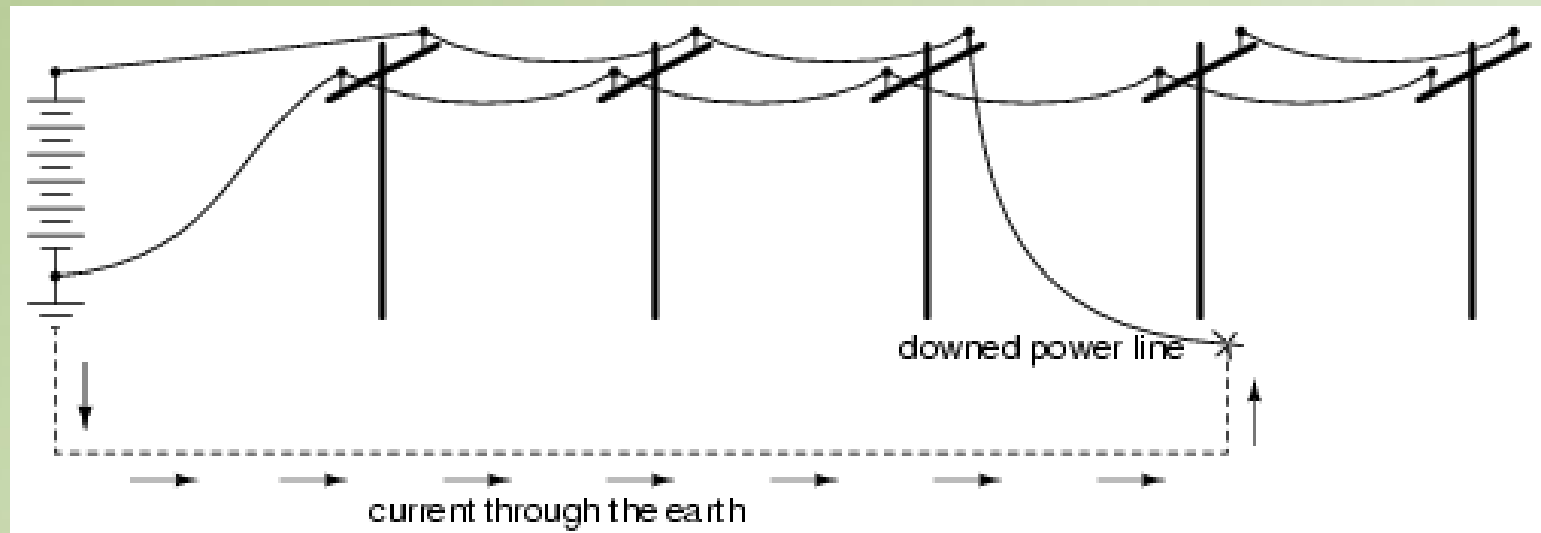
Safe Practices

- If at all possible, **shut off the power** to a circuit before performing any work on it
 - Called Zero Energy State

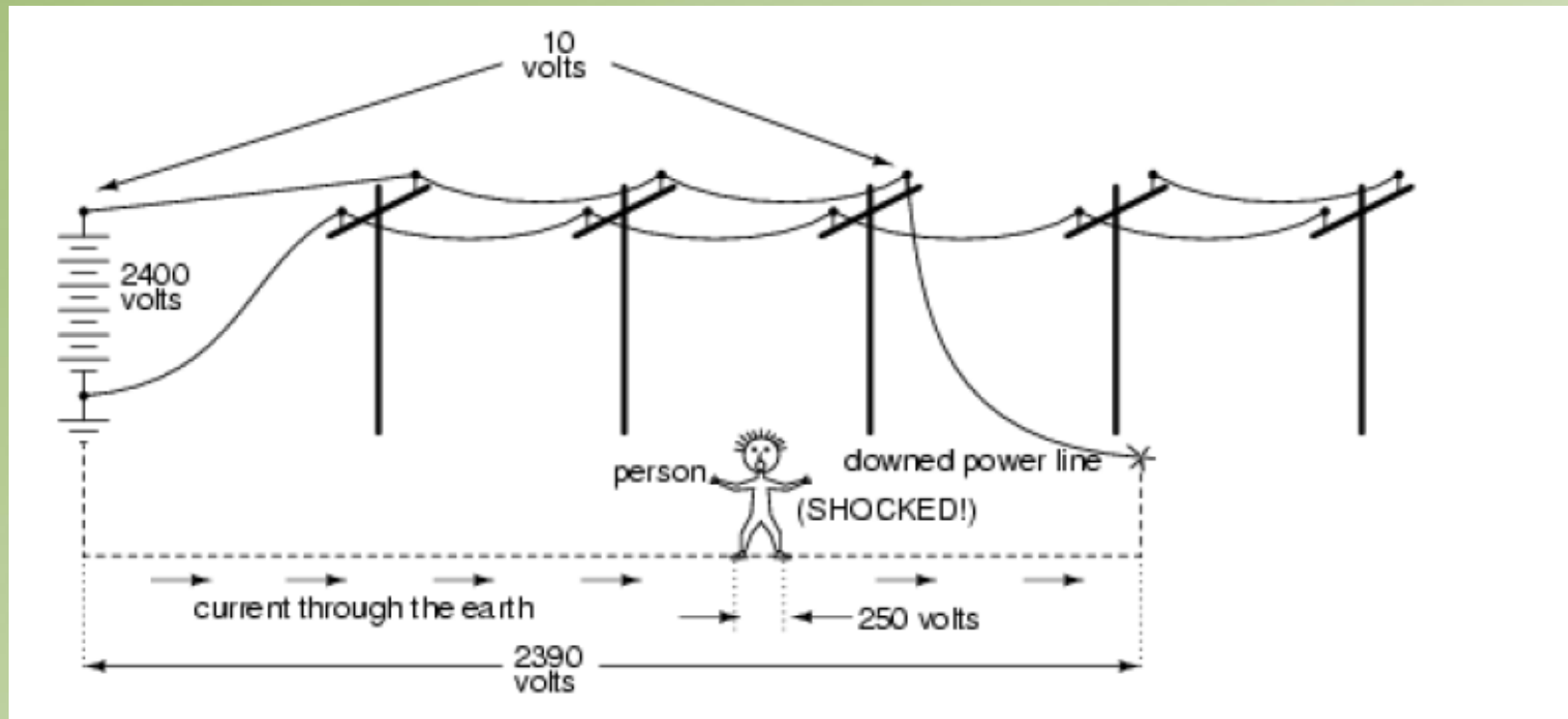


Safe Practices

- **Skin/body resistance** have a lot to do with the relative hazard of electric circuits → **keep them dry**
- In the household, the bathroom is one of the more likely places where shock hazard is a definite threat
- **Downed lines:** What to do if you encounter one?



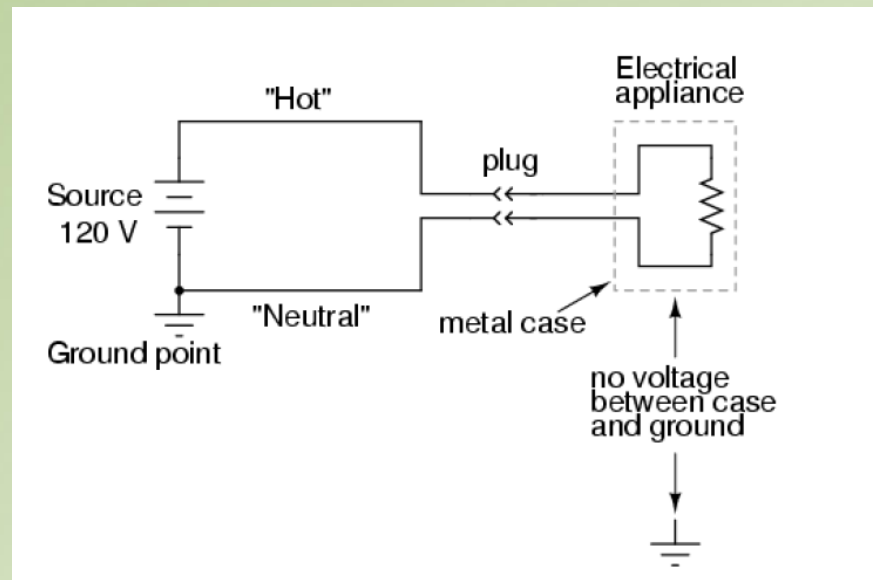
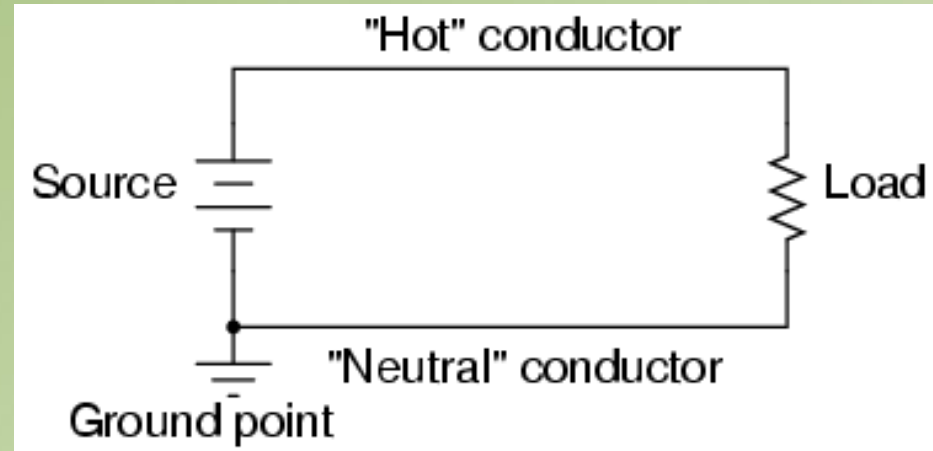
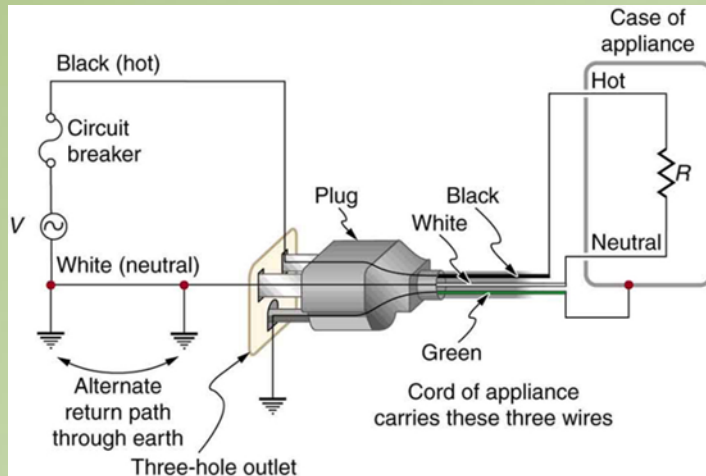
Grounded power lines



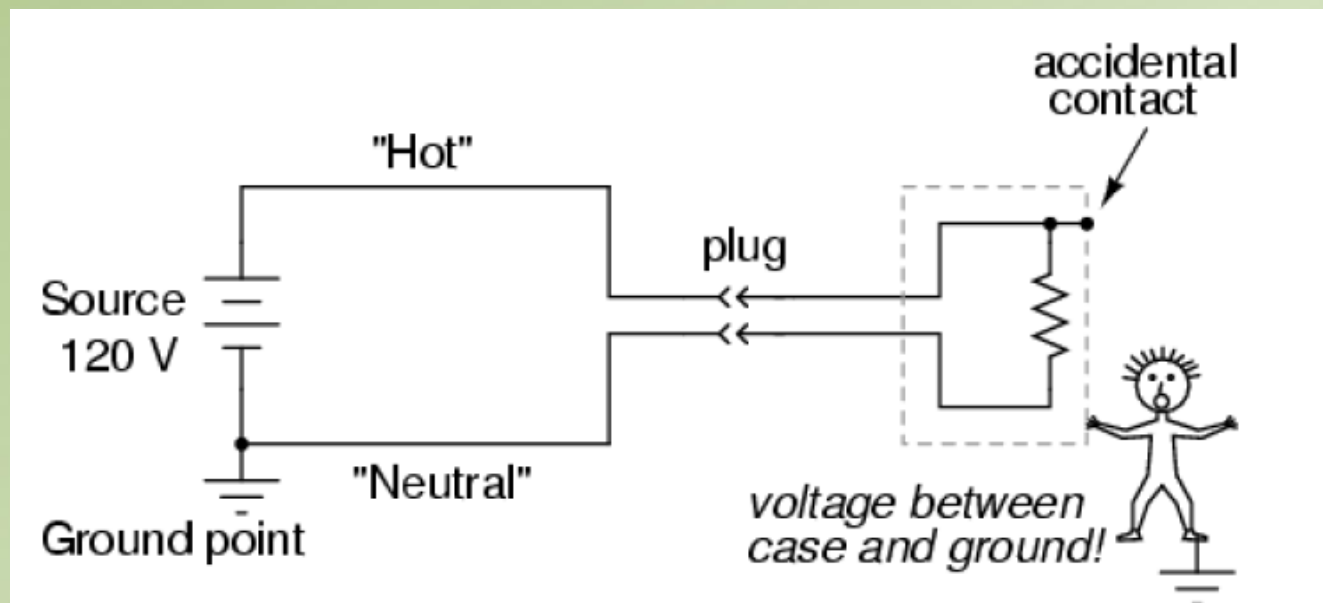
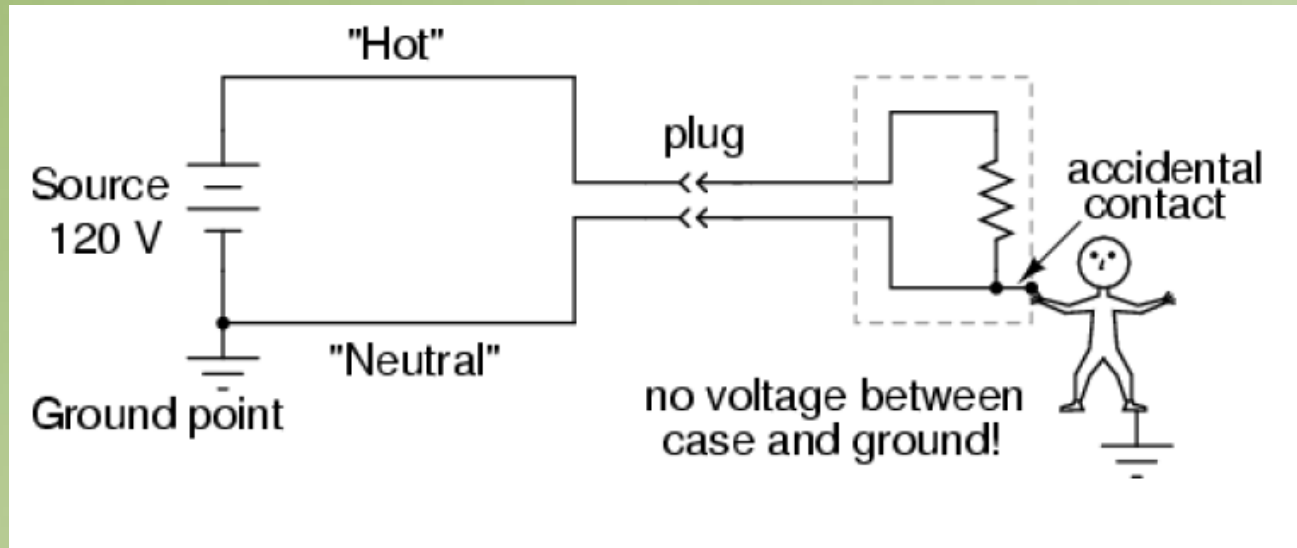
Safe Circuit Design/Practices

In a typical household appliance:

- Hot wire
- Neutral wire
- Ground

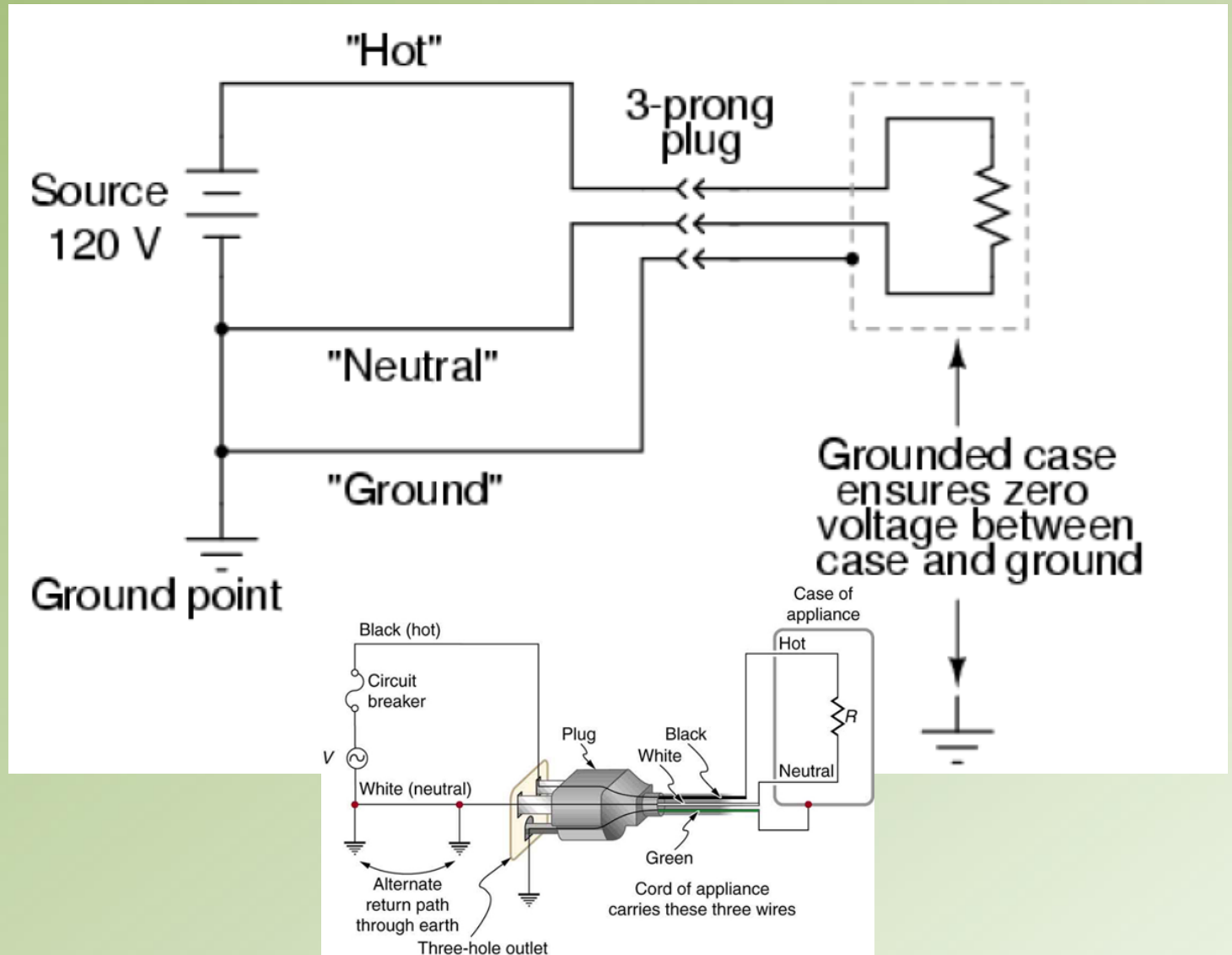


Safe Circuit Design/Practices



Safe Circuit Design/Practices

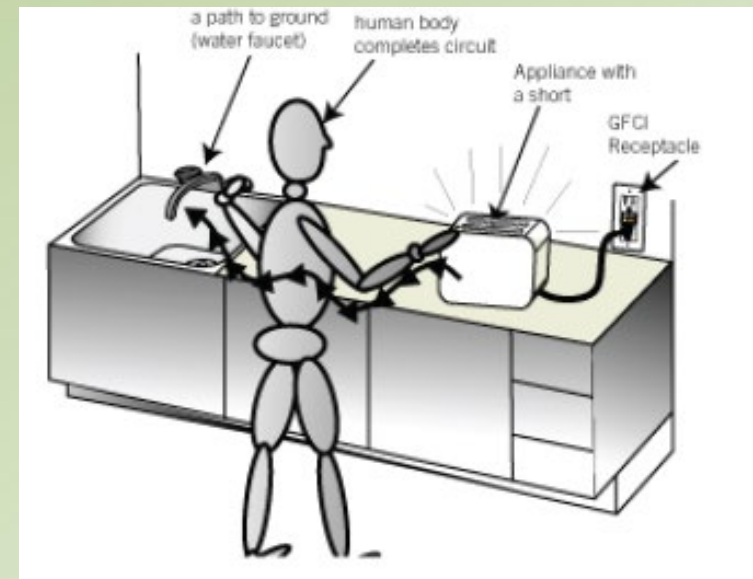
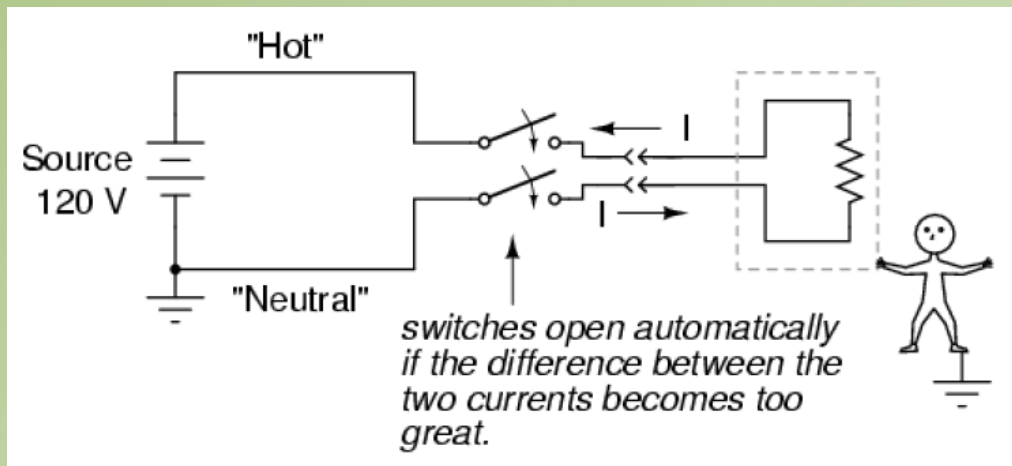
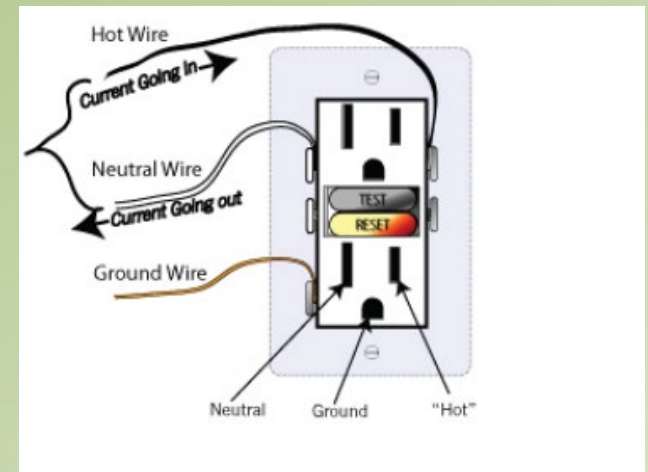
Safety ground protects user in case of accidental contacts



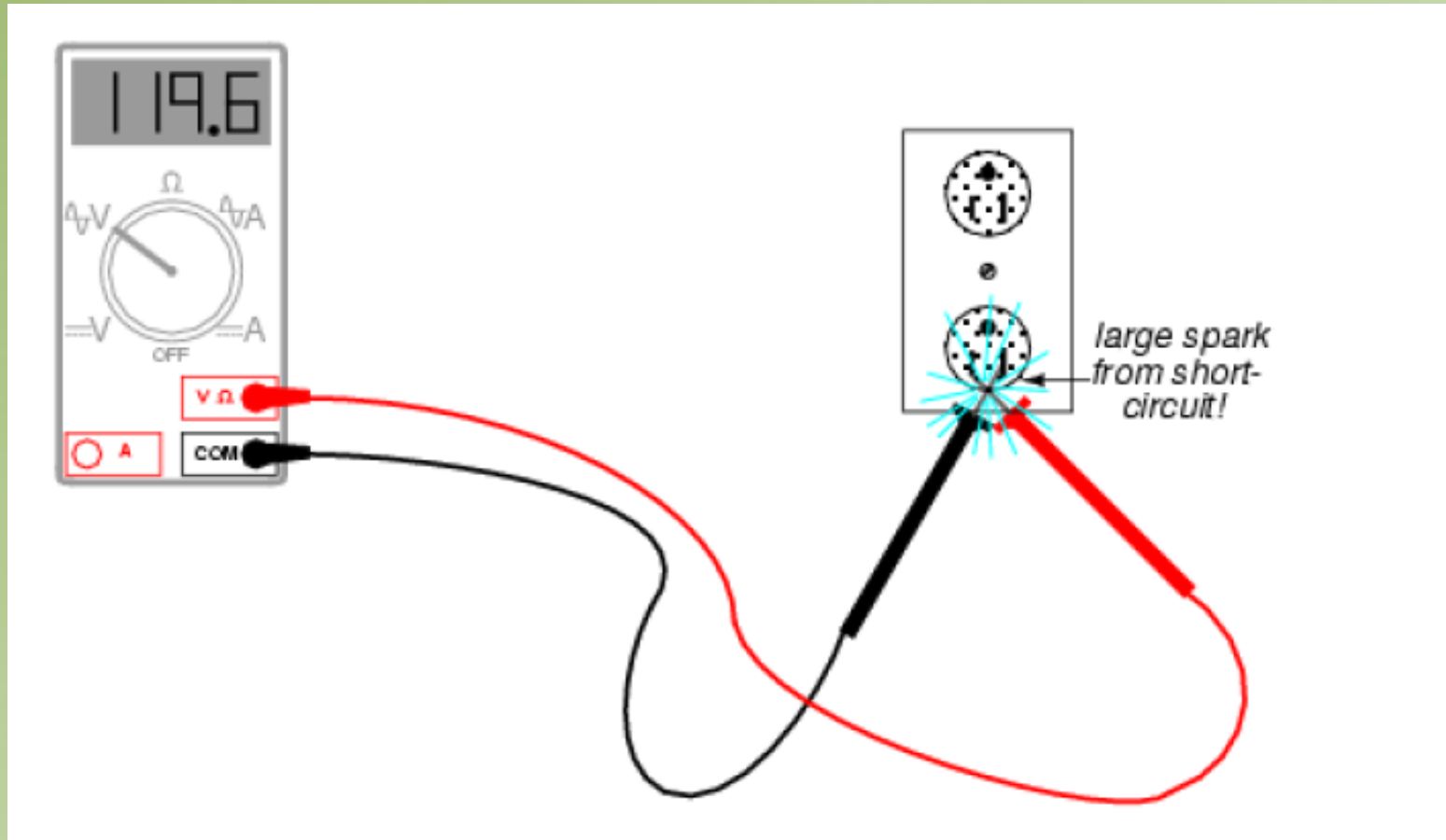
Safe Circuit Design/Practices

Ground Fault Current Interrupter, or GFCI:

- Swimming pools, sauna, bathrooms, wet environments

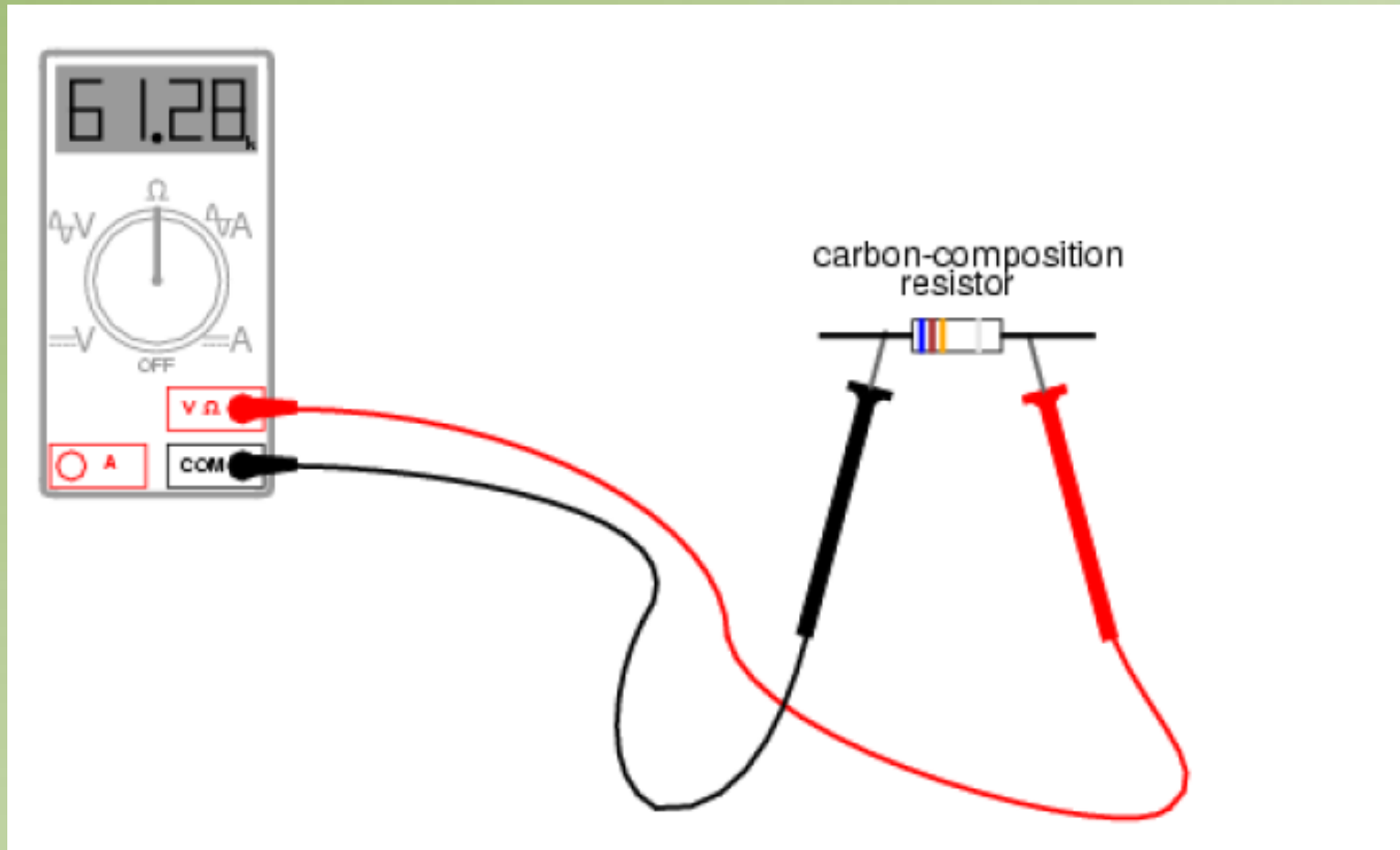


Safe Use of Equipment



Be careful not to make short circuits when measuring high voltages with multi-meters

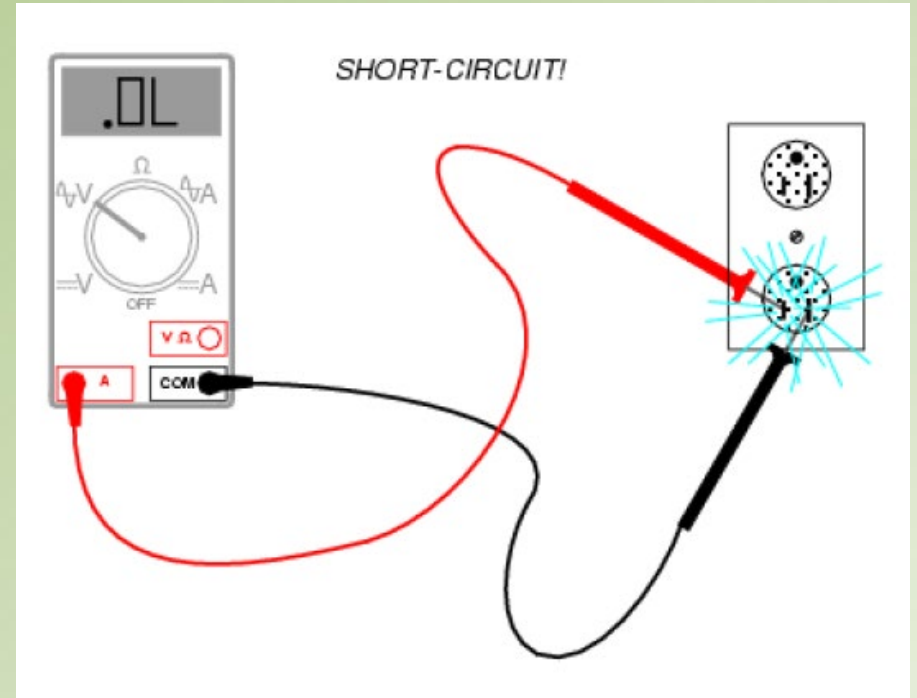
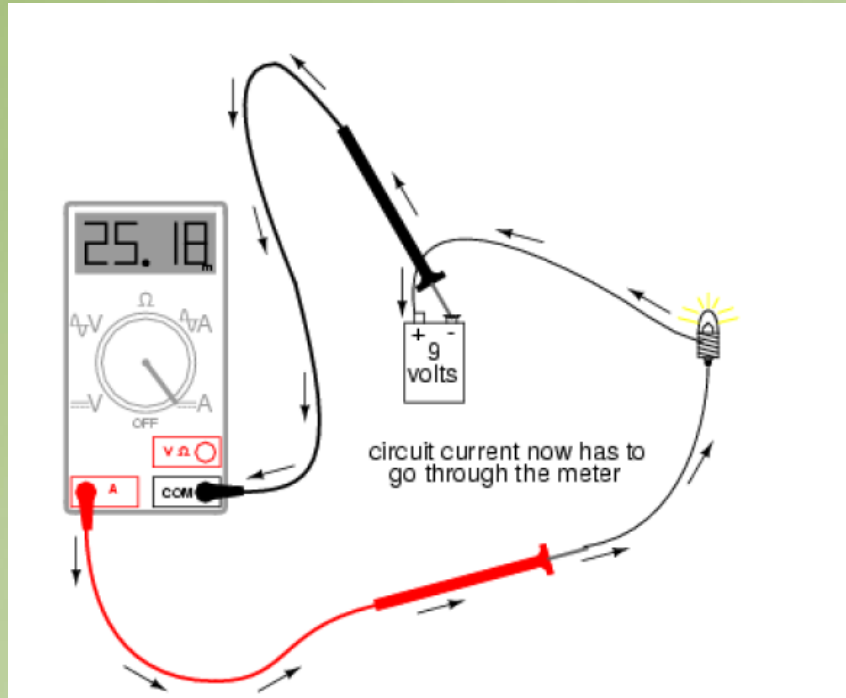
Safe Use of Equipment

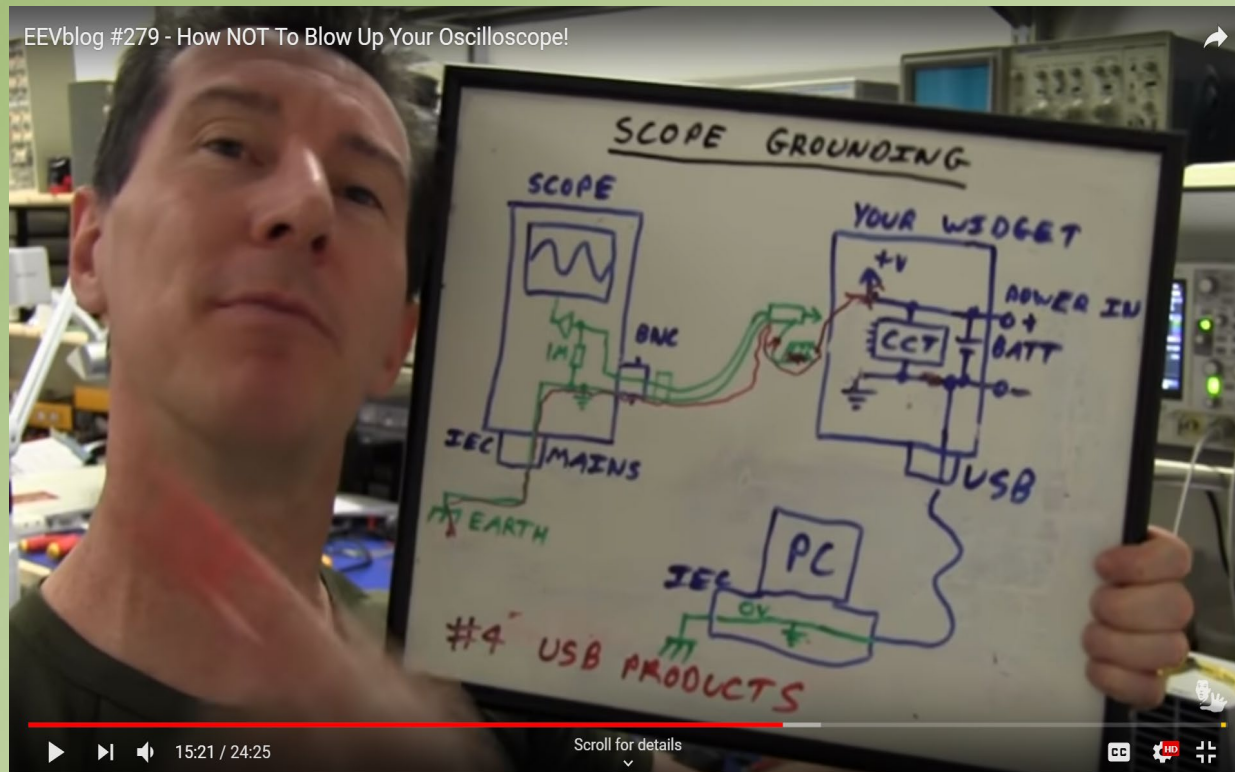


De-energize the circuit when making resistance measurements

Safe Use of Equipment

- The most hazardous and complex application of the multimeter is in the measurement of current.



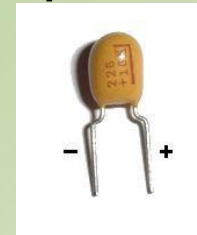


- Beware of the issue → e.g., when doing experiments on H-bridge (e.g., measuring voltage across motor using scope)

Safety for electronic circuits

Polarity problems may result in fires and flying particles:

- Electrolytic capacitors
 - If polarized incorrectly, they may explode!
- Tantalum capacitors (same)
- Semiconductors
 - If polarized incorrectly, they may burn up in the circuit.



- Inductors can generate high voltages even in low voltage electronic circuits

Recap

Which statement(s) is (are) true?

- (a) Electric current can burn a tissue
- (b) A main effect of electric current flow in human body is on the nervous system
- (c) Voltage can be dangerous due to the current it may cause
- (d) All the above

Which statement(s) is (are) true?

- (a) AC voltage at power line frequency is less dangerous than DC
- (b) If a circuit is operating with voltages normally below 30V, we should still worry about safety issues due to high voltages
- (c) Leather shoes can provide a good degree of safety against electric shock
- (d) All the above

In a power line that is not grounded:

- (a) The circuit is unpredictable in terms of safety
- (b) Can touch the circuit by both hands without any problem
- (c) Accidental grounds cannot happen
- (d) None of the above

An Ohmmeter is used to measure a $10\text{ k}\Omega$ resistance (with 10% tolerance) in a circuit while it is powered. Indicate which statement is true:

- (a) The ohmmeter's reading is between $9.9\text{ k}\Omega$ to $10.1\text{ k}\Omega$
- (b) The reading will have a gross error
- (c) The Ohmmeter may be damaged
- (d) Either (b) or (c) can happen